

Question 1: Answer the following (MCQ) questions

1- check bit errors.

**A. nodal processing** B. queueing delay C. transmission delay D. propagation delay

2- determine output link.

**A. nodal processing** B. queueing delay C. transmission delay D. propagation delay

3- time waiting at output link for transmission.

A. nodal processing **B. queueing delay** C. transmission delay D. propagation delay

4- depends on congestion level of router.

A. nodal processing **B. queueing delay** C. transmission delay D. propagation delay

5-  $= L/R$ ,  $L$ : packet length (bits),  $R$ : link *bandwidth* (bps).

A. nodal processing B. queueing delay **C. transmission delay** D. propagation delay

6-  $= d/s$ ,  $d$ : length of physical link,  $s$ : propagation speed in medium.

A. nodal processing B. queueing delay C. transmission delay **D. propagation delay**

7- self-replicating infection by receiving/executing object (e.g., e-mail attachment)

**A. virus** B. worm C. spyware D. botnet

8- self-replicating infection by passively receiving object that gets itself executed

A. virus **B. worm** C. spyware D. botnet

9- malware can record keystrokes, web sites visited, upload info to collection site.

A. virus B. worm **C. spyware** D. botnet

10- attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic.

A. virus B. worm C. spyware **D. DoS**

Question 2: Mark false or true

1- RFC stands for Request for comments. (true)

2- IETF stands for Internet Engineering Task Force. (true)

3- protocols define format, order of msgs sent and received among network entities, and actions taken on msg transmission, receipt. (true)

4- voice over DSL phone line goes to Internet. (false)

5- data over DSL phone line goes to telephone net. (false)

6- In DSL, voice, data transmitted at different frequencies over **dedicated** line to central office. (true)

7- DSL stands for digital subscriber line. (true)

8- In cable network, data, TV transmitted at different frequencies over **shared** cable distribution network. (true)

9- link transmission rate, aka link *capacity*, aka *link bandwidth*. (true)

10- packet transmission delay = time needed to transmit  $L$ -bit packet into link . (true)

Question 2: Mark false or true.(20 marks)

11- *twisted pair (TP)* is two insulated copper wires. (true)

12- In packet-switching hosts break application-layer messages into *packets*. (true)

13- If arrival rate (in bits) to link exceeds transmission rate of link for a period of time: packets will queue, wait to be transmitted on link . (true)

14- If arrival rate (in bits) to link exceeds transmission rate of link for a period of time: packets can be dropped (lost) if memory (buffer) fills up. (true)

15- *forwarding*: determines source-destination route taken by packets. (false)

16- *routing*: move packets from router's input to appropriate router output. (false)

17- connecting each access ISP to each other directly *does scale*. (false)

18-  $R$ : link bandwidth (bps),  $L$ : packet length (bits),  $a$ : average packet arrival rate,  $La/R \sim 0$ : avg. queueing delay small. (true)

19-  $R$ : link bandwidth (bps),  $L$ : packet length (bits),  $a$ : average packet arrival rate,  $La/R \rightarrow 1$ : avg. queueing delay large. (true)

20-  $R$ : link bandwidth (bps),  $L$ : packet length (bits),  $a$ : average packet arrival rate,  $La/R > 1$ : more "work" arriving than can be serviced, average delay infinite! (true)

21- lost packet may be retransmitted by previous node, by source end system, or not at all. (true)

22-*throughput*: rate (bits/time unit) at which bits transferred between sender/receiver. (true)

23-In ISO/OSI reference model, *presentation*: allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions. (true)

24-Internet stack missing the layers *presentation and session*. (true)

25- infected host can be enrolled in botnet, used for spam. (true)

26- *IP spoofing*: send packet with true source address. (false)

27- server not always-on host. (false)

28- server has permanent IP address. (true)

29- In client-server architecture, clients communicate with server. (true)

30- In client-server architecture, clients may be intermittently connected. (true)